

more technical definition of software:

Score	Requirement	Technical Conditions
0.2	arms approach the beaker	both grippers are within a specified distance of the gripper
0.4	beaker's grasped	beaker is held in the grippers for min. duration (what about beaker distance off the ground?)
0.6	beaker's lifted	beaker rises 3 cm above table
0.8	beaker reaches target	beaker center enters target region
1.0	beaker's released correctly	beaker remains in target after grippers open

goal: place the beaker on a measuring scale

b/c the measuring scale is involved which means that the stages need to be broken down further to completely encompass the intricacies of successfully performing that action.

is this constitutes under the lift condition.

1. The easiest first implementation? manual scoring

- record every eval rollout
- watch front and top camera views
- mark farthest completed stage

Q: What happens when the policy fails?

to fix this have to also annotate the actions that took place

↳ this doesn't really make sense

Other Questions to Consider:

- does marking the locations where the bins are going to be located w/ physical tape make/reduce the policy unable to complete the task b/c it learns the surroundings rather than the confines of the robot's task?

more reasonable workflow

- manually score 5-10 trials:
 - evaluate how reasonable the

Still need to decide but this can be decided after recording the demonstrations for the arms.

Plan for first initial test:

↳ yep this is a problem so can make a template that can be placed on table to indicate the positions. (Nandini can work on creating this template tomorrow)

- start w/ one camera (even though we have 3 cameras available & eventually are planning on having a top-down camera)

- Note! want to use "bins" of locations where the exact positions within the bins vary to create a dataset best suited to the strengths of ACT policy training.

Q: do we then want to order a 4th camera for top down view? assuming that if we have side view cameras we would want the cameras to capture both sides.

- Start by using a small pilot dataset:

problem observed in policy	implication/suggestion
confuses front & back positions	add top camera
beaker becomes hidden behind arm	add top or side view
reaches correctly but can't lift	more likely a data/grasp issue
works only on trained positions	improve position coverage
can't identify target area	improve camera angle / add top view

Q: is it possible to record this data in simulation to answer these questions faster?

↳ not really applicable b/c the goal is to have ACT work on our specific setup not to prove in general that ACT works.

Q: plan for ensuring operator is controlling arms w/ only the info the policy is being trained w/ → so only seeing the camera frame

↳ curious as to why the position data doesn't matter? b/c pol can't make sense of that information at the same rate as the policy? but that's also true of the camera frames?

long term goals?

Some suggestions that have been discussed:

- haptic info: using impedance + current draw to learn dynamics into & translates that back into resistance for the gripper.